

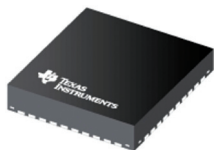
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Wi-Fi 6 companion ICs

Texas Instruments' SimpleLink family of Wi-Fi 6 companion ICs is designed for affordable Wi-Fi connections in high-density or high-temperature environments up to 105°C. The CC33xx family includes devices for Wi-Fi 6



only or Wi-Fi 6 and Bluetooth Low Energy 5.3 connectivity in a single IC. These

devices enable secure IoT connections with reliable RF performance in industrial markets such as grid infrastructure, medical, and building automation. The CC33xx devices provide efficient Wi-Fi network performance, stable connections across multiple access points, and operate in a temperature range of -40°C to 105°C. They support OFDMA technology and BSS colouring for enhanced network performance and support WPA3 security features. The devices can be easily attached to TI and other companies' MCUs and processors, including AI-capable processors and MCUs supporting various protocols like Thread, Zigbee, and Matter.

Texas Instruments
<https://www.ti.com>

MEMS water-proof absolute pressure sensor

STMicroelectronics has launched the market's first MEMS water/liquid-proof absolute pressure sensor, the ILPS28QSW, designed for industrial applications. The sensor offers a 10-year longevity program and is suitable for challenging indoor and outdoor environments. It comes in a sealed,

cylindrical, surface-mountable package with a ceramic substrate that provides high resistance to liquid permeability. The sensor has an IP58 ingress-protection rating, allowing it to withstand immersion in over one me-



tre of water. It can sustain up to 10Bar over-pressure. The ILPS28QSW

provides accurate absolute pressure readings, selectable full-scale ranges of 260-1260hPa and 260-4060hPa, and operates in a wide temperature range of -40°C to 105°C. It is ideal for gas and water metering equipment, weather monitors, air-conditioning smart filters, and home appliances. The sensor features ST's Qvar electrostatic charge sensing channel, enabling liquid leakage detection and other value-added features. It has low operating current, integrated digital features, and a selectable output data rate.

STMicroelectronics
<https://www.st.com>

Electromechanical relay switches

Pasternack has introduced a series of ruggedised electromechanical relay switches designed to ensure reliable



RF signal routing across a wide range of frequency bandwidths, from DC to 40GHz. These

switches comply with IP64 standards and feature a Level 1 moisture seal, making them suitable for outdoor environments and testing conditions

involving moisture exposure. The relay switches offer low insertion loss (as low as 0.15dB) and high isolation levels of up to 90dB, ensuring exceptional signal integrity and minimising signal degradation. The switches can handle power levels of up to 160 watts continuous wave (CW) at 1MHz, providing consistent and reliable performance. Designed in compact packages adhering to MIL-grade standards, the switches feature SMA/N connectors and 50-ohm impedance. Key specifications include maximum VSWR of 1.4:1, minimum frequencies of 6GHz or 1GHz, maximum frequencies of 18GHz or 12GHz, maximum insertion loss of 0.4dB or 0.5dB, and minimum isolation of 65dB or 60dB.

Pasternack
<https://www.pasternack.com>

Compact photorelay

Toshiba Electronic Devices & Storage Corporation has introduced its latest photorelay, the TLP3476S, offering significant advancements over its predecessor. Housed in a compact



S-VSON4T package, the TLP3476S features a reduced turn-on time

that is half as long as the previous model, making it ideal for applications in semiconductor testers and pin electronics that require shorter switching times. The photo relay delivers faster operating speed and improved optical coupling efficiency. With a 20% slimmer design, it is suitable for equipment requiring multiple boards, contributing to overall size reduction. The TLP3476S provides engineers and designers with enhanced flexibility and performance for their applications. The photorelay incorporates im-